



Effect of Preharvest Application of Melatonin for Maintaining Postharvest Quality and Shelf Life in Guava

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ABSTRACT

Background: Guava (*Psidium guajava* L.) is a highly perishable climacteric fruit characterized by a rapid ripening process and a short post-harvest life, which leads to significant economic losses. Melatonin has emerged as a potent antioxidant and signaling molecule capable of delaying senescence and preserving fruit quality.

Methods: A field experiment was conducted using a Randomized Block Design (RBD) with four replications. Guava trees were subjected to five preharvest treatments: T₁ (Control), T₂ (50 µM), T₃ (100 µM), T₄ (150 µM) and T₅ (200 µM) melatonin. The treatments were applied as foliar sprays two weeks prior to the anticipated harvest date.

Result: Among the concentrations tested, fruits treated with 100 µM melatonin maintained significantly higher firmness, titratable acidity, ascorbic acid (Vitamin C) and total phenolic content compared to the control group. Furthermore, this treatment effectively minimized physiological loss in weight (PLW) and extended the shelf life of the fruits up to 9 (8.73) days under ambient conditions.

Key words: Guava, Melatonin, Post-harvest quality, Pre-harvest, Shelf life.

INTRODUCTION

Guava is a climacteric fruit enriched with bioactive compounds, yet its commercial utilization is constrained by rapid post-harvest deterioration driven by accelerated physicochemical processes. Such degradation contributes to pronounced quality losses and considerable economic implications across the supply chain (Labib *et al.*, 2025). The crop exhibits seasonal flowering patterns influenced by climatic conditions, resulting in multiple harvests in southern India. However, fruits obtained during the rainy season are often of inferior quality compared to winter harvests, largely due to higher pest incidence and reduced storage potential (Sahu *et al.*, 2025; Sharma, 2024). As a climacteric fruit, guava undergoes sharp increases in respiration and ethylene biosynthesis following harvest, which hastens ripening and senescence. These processes manifest as softening, loss of firmness, decline in nutritional attributes and substantial post-harvest losses (Yadav *et al.*, 2022). Consequently, the development of effective strategies to delay ripening and preserve quality is critical for extending shelf life and ensuring market sustainability.

Melatonin (N-acetyl-5-methoxytryptamine) is a versatile signalling molecule that occurs ubiquitously across diverse plant taxa (Ze *et al.*, 2021). In recent years, the use of eco-friendly natural compounds or bio stimulants, such as melatonin, has gained importance due to their ability to regulate biosynthetic pathways involved in the formation of key quality attributes of fruits during ripening and subsequent post-harvest storage (Mubarak *et al.*, 2023). Melatonin has emerged as a promising bio regulator owing to its functions in plant growth regulation, enhancement of stress tolerance and modulation of fruit ripening (Gao *et al.*,

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2022). Exogenous application of melatonin has been reported to delay fruit ripening, suppress ethylene biosynthesis and strengthen antioxidant defence mechanisms, thereby prolonging shelf life in several fruit species (Onik *et al.*, 2021). Melatonin exhibits antioxidant activity during postharvest storage, thereby contributing to the extension of shelf life in fruits and vegetables (Gurjar *et al.*, 2022). Preharvest application of melatonin has been shown to delay climacteric ethylene peaks, thereby

maintaining fruit firmness and modulating chlorophyll-carotenoid transitions, which collectively enhance post-harvest quality across diverse fruit species (Aghdam and Arnao, 2024). Multiple investigations have explored the role of melatonin in regulating fruit yield and quality (Xie *et al.*, 2022a). Although melatonin has been widely studied for its regulatory roles in plants, comprehensive evaluations of its preharvest application in guava are lacking, underscoring the necessity for targeted investigations to optimize post-harvest quality and shelf life. Therefore, this study was conducted to assess the impact of melatonin on postharvest quality attributes and shelf life of guava fruits.

MATERIALS AND METHODS

The present investigation was carried out in the fruit orchard by the Department of Fruit Science, SRM College of Agricultural Sciences, Chengalpattu, Tamil Nadu, India, during 2025-2026. The experimental site is situated at 12°23' 19.7" N latitude and 79°44' 37.4" E longitude, with an elevation of approximately 50 m above the mean sea level. The region experiences a tropical climate, with ambient temperatures ranging between 23°C and 38°C. The soil at the site is predominantly clayey, characterized by moderate fertility and adequate drainage capacity, making it suitable for fruit crop cultivation.

Healthy guava trees were selected for the study and the experiment was conducted in a Randomized Block Design (RBD) with four replications. Five treatments were evaluated: T₁ (control, water spray), T₂ (melatonin 50 µM), T₃ (melatonin 100 µM), T₄ (melatonin 150 µM) and T₅

(melatonin 200 µM). Melatonin solutions were freshly prepared by dissolving the required amount in a small volume of ethanol, followed by dilution with distilled water to obtain the desired concentrations (50-200 µM). Foliar sprays were applied uniformly two weeks before harvest using a battery-operated sprayer during early morning hours to ensure effective absorption and minimize evaporation. Fruits were harvested at physiological maturity and transported carefully to the laboratory for analysis (Dangwal *et al.*, 2026). All laboratory analyses were conducted according to the standardized procedures outlined in Table 1.

The recorded data were statistically analysed by using Analysis of Variance (ANOVA) appropriate for RBD to determine the significance of treatment effects with the help of the KAUGRAPES web-based statistical analysis software (Gopinath *et al.*, 2021).

RESULT S AND DISCUSSION

Fruit firmness declined significantly as storage progressed, regardless of treatment. Fruit softening during storage mainly occurs due to the enzymatic breakdown of cell wall constituents such as pectin, cellulose and hemicellulose, mediated by enzymes like polygalacturonase and pectin methyl esterase (Kucuker *et al.*, 2023). A gradual decline in fruit firmness was observed across all treatments throughout the storage period, indicating progressive softening during storage. However, fruits treated with 100 µM melatonin retained greater firmness, whereas the untreated controls exhibited the lowest firmness values (Table 2). Earlier studies have indicated that melatonin

Table 1: Parameters recorded during the study in guava fruit.

Parameter	Methodology	Units	Reference
Fruit firmness	Penetrometer	Newtons (N).	(Abbott, 1999)
Total soluble solids	Refractometer	°Brix	(AOAC, 2005)
Titrateable acidity	Acid-base titration method	Per cent %)	(AOAC, 2005)
Ascorbic acid	Indophenol method.	mg/100g	(AOAC, 2005)
Total phenolic content	Folin Ciocalteu colorimetric method	GAE/100g	(Singleton and Rossi, 1965)
Total sugars	Phenol method	Per cent %)	(DuBois <i>et al.</i> , 1956)
Reducing sugar	DNS reagent method	Per cent %)	(Miller, 1959)
Physiological loss in weight	$PLW (\%) = \frac{(W_0 - W_1)}{W_0} \times 100$	Per cent %)	Wills and Golding (2016)

Table 2: Effect of preharvest melatonin on firmness and TSS content of guava.

	Firmness (N)			TSS (° Brix)		
	Day 3	Day 6	Day 9	Day 3	Day 6	Day 9
T ₁ - Control	6.51	5.23	4.14	12.19	12.84	13.56
T ₂ - 50 µM	7.12	5.86	4.77	10.81	12.23	13.06
T ₃ - 100 µM	8.09	7.25	6.91	10.54	11.76	12.62
T ₄ - 150 µM	7.63	6.56	5.59	11.07	12.68	13.53
T ₅ - 200 µM	7.17	6.34	5.38	10.67	12.52	13.31
SE(d)	0.11	0.09	0.06	0.17	0.14	0.23
CD (≤0.05)	0.23	0.20	0.13	0.36	0.31	0.51

can inhibit the activity of cell wall degrading enzymes, thereby preserve cell wall integrity and delaying softening in papaya fruit (Borthakur *et al.*, 2024). Thus, the improved firmness observed in melatonin treated fruits may be attributed to better preservation of structural polysaccharides and enhanced membrane stability during storage (Garrido Auñón *et al.*, 2025).

Total soluble solids (TSS) serve as an indicator of fruit maturation or progression into advanced storage stages (Kasilingam *et al.*, 2025). Data presented in Table 2 indicate significant differences in the effects of melatonin on fruit TSS content. The progression of ripening during storage was reflected by gradual increase in TSS as storage progressed across all treatment. Among the treatments, fruits treated with 100 μM of melatonin recorded the lowest TSS, while the control fruits exhibited the higher values. The rise in TSS during storage is a typical ripening response, mainly resulting from the breakdown of complex carbohydrates into soluble sugars. Application of melatonin (MT) effectively delays mango ripening by inhibiting starch to sugar conversion, maintaining cell wall integrity, modulating antioxidant metabolism and regulating the biosynthesis of ethylene and abscisic acid (Njie *et al.*, 2022). Similar result was observed in banana (Devi *et al.*, 2025) and Japanese plum (Cortes-Montana *et al.*, 2023).

The decline in titratable acidity of guava fruits during storage was markedly reduced by exogenous melatonin treatment. The observed decline in titratable acidity (TA) during storage can be attributed to the metabolism of organic acids within the fruit or their breakdown through respiratory activity (Wang *et al.*, 2022 and Jiang *et al.*, 2023). However, fruits treated with melatonin retained significantly higher acidity compared to other treatments. Among the treatments, 100 μM recorded the highest acidity values, indicating the effectiveness of melatonin in delaying the decline of organic acids during storage (Fig 1). Melatonin application may have moderated the rate of respiration and related metabolic activities, thereby reducing the rapid consumption of organic acids. Similar findings have been reported where melatonin treatment helped maintain higher acidity levels in fruits by slowing ripening related biochemical changes in Japanese plum (Cortes-Montana *et al.*, 2023).

Reduced sugar accumulation is a characteristic marker of delayed postharvest ripening and senescence, primarily associated with the inhibition of amylase and phosphorylase activities (Hanif *et al.*, 2020). During storage, complex carbohydrates were converted into soluble sugars (Johari *et al.*, 2023). Application of melatonin increased the total sugar compared with the control at the end of

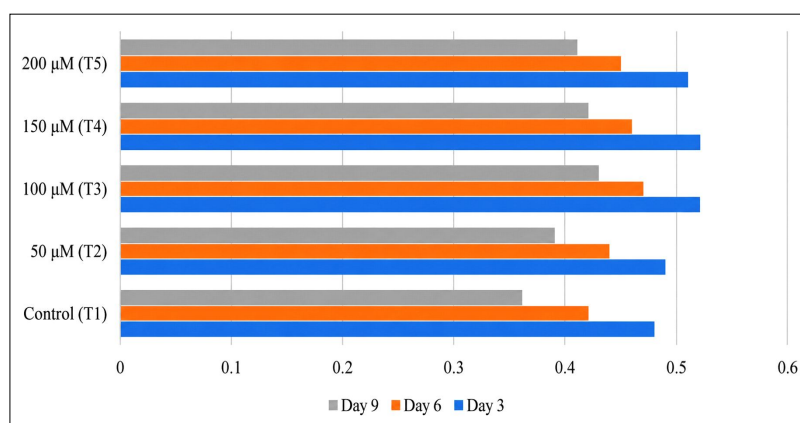


Fig 1: Effects of preharvest melatonin on titratable acidity of guava.

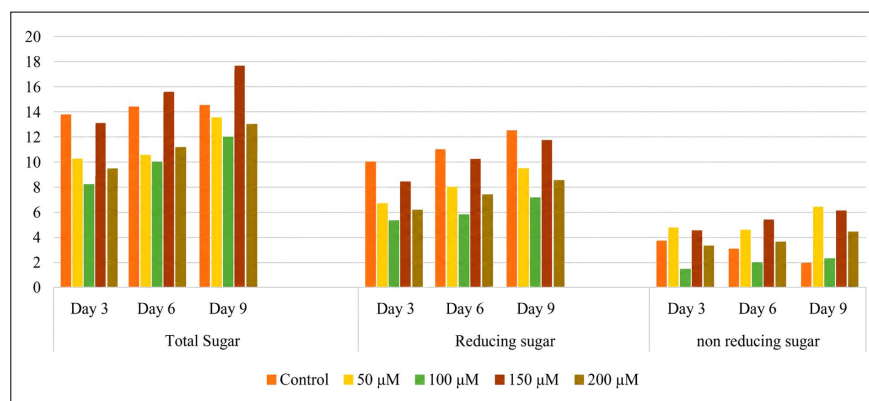


Fig 2: Effects of preharvest melatonin on total, reducing and non-reducing sugars of guava.

storage period indicating slower metabolic activity (Fig 2). Total sugar content showed a steady increase throughout the storage in all treatments, reflecting the normal ripening process. The reduced accumulation of sugars suggests that melatonin regulated carbohydrate metabolism and delayed the breakdown of polysaccharides into simple sugars, thereby slowing the ripening process. Similar findings were reported by Zhao *et al.* (2023), who observed that melatonin treatment can regulate sugar metabolism by influencing enzymes involved in carbohydrate transformation in pear fruits. Thus, the slower increase in total sugars in melatonin treated fruits indicates delayed ripening and better preservation of fruit quality during storage.

The contents of reducing and non-reducing sugars in guava fruits exhibited gradual changes throughout the storage period. At the end of the storage period, melatonin treated fruits showed considerably lower levels of both reducing and non-reducing sugars compared with the control fruits (Fig 2). This effect may be attributed to melatonin's inhibition of ethylene biosynthesis and the concentration-dependent suppression of starch conversion to soluble sugars, thereby contributing to delayed ripening and the maintenance of fruit quality (Liu *et al.*, 2020). Similar results were observed in papaya (Wang *et al.*, 2022) and peach (Zhou *et al.*, 2023).

Ascorbic acid constitutes a fundamental antioxidant that governs the progression of fruit ripening by modulating reactive oxygen species metabolism and maintaining the intracellular redox equilibrium (Arabia *et al.*, 2024). In this study, a consistent decline in ascorbic acid content was observed across all treatments throughout the storage period, reflecting the gradual degradation of vitamin C as ripening progressed. Among the treatments, fruit treated with 100 μM melatonin retained significantly higher ascorbic acid content, while the control fruits exhibited the lowest values (Fig 3). This effect may be attributed to the application of melatonin, which enhances the accumulation of metabolites during fruit ripening by elevating total flavonoid and phenolic contents as well as ascorbic acid levels (Fan *et al.*, 2022a). Numerous investigations have demonstrated

that melatonin significantly modulates fruit metabolite profiles by enhancing and sustaining levels of ascorbic acid (Zhang *et al.*, 2020). Comparable outcomes were reported in guava (Fan *et al.*, 2022b), kiwifruit (Luo *et al.*, 2022) and pear (Liu *et al.*, 2024).

Melatonin is widely recognized as a potent antioxidant that limits free radical accumulation in plant tissues while supporting the preservation of non-enzymatic antioxidant metabolites, including anthocyanins, phenolics and flavonoids (Xie *et al.*, 2022b). In present investigation, the total phenolic content declined progressively during storage irrespective of melatonin treatment; however, the reduction was more pronounced in control fruits compared to melatonin treated fruits. By the end of storage period, fruits treated with 100 μM melatonin exhibited the highest phenolic content (Fig 3). Comparable increases in phenolic compound concentrations following melatonin application have recently been reported in table grapes (Wang *et al.*, 2020), litchi (Marak *et al.*, 2024) and raspberry (Rahmanzadeh-Ishkeh *et al.*, 2024).

The percentage of physiological weight loss (PLW) in guava fruit exhibited a progressive increase throughout the storage period. At the end of storage, fruits treated with 100 μM melatonin exhibited the lowest percentage of physiological weight loss (PLW), whereas the untreated controls showed significantly higher weight loss (Table 3). Exogenous melatonin has been reported to stimulate

Table 3: Effect of preharvest Melatonin on PLW and shelf life of guava.

	PLW (%)			Shelf life
	Day 3	Day 6	Day 9	
T ₁ - Control	3.32	7.19	11.78	5.21
T ₂ - 50 μM	2.83	6.39	10.45	6.18
T ₃ - 100 μM	1.92	4.47	8.23	8.73
T ₄ - 150 μM	2.21	5.17	9.04	7.36
T ₅ - 200 μM	2.52	5.84	9.61	6.79
SE(d)	0.04	0.04	0.10	0.11
CD (≤ 0.05)	0.09	0.08	0.22	0.22

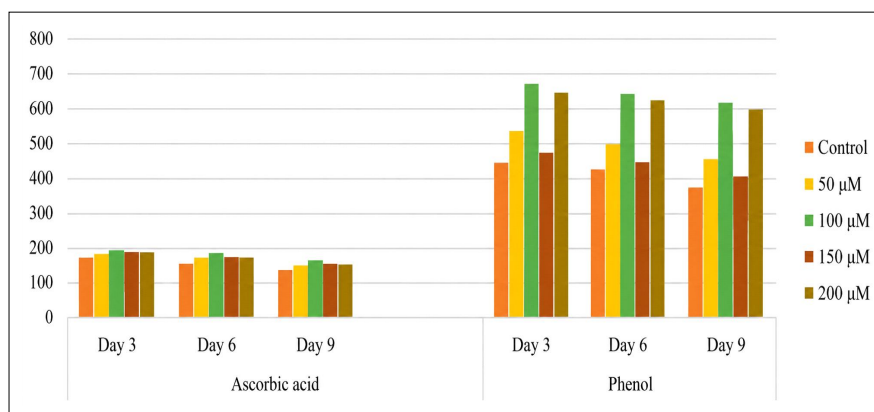


Fig 3: Effects of preharvest melatonin on ascorbic acid and phenol of guava.

endogenous melatonin biosynthesis, thereby attenuating PLW (Li *et al.*, 2023). The mitigation of weight loss following melatonin application may be attributed to the downregulation of metabolic activity, resulting in reduced energy expenditure and diminished rates of transpiration and respiration in stored fruit (Padilla-González *et al.*, 2026; Bal, 2021). Concordant findings have been documented in other fruit species, including plum (Zhang *et al.*, 2024), litchi (Xie *et al.*, 2022b) and banana (Anchana *et al.*, 2023).

Exogenous melatonin acts as an effective natural preservative that enhances postharvest fruit shelf life by mitigating physiological processes related to oxidative damage and strengthening antioxidant defences (Budiarto *et al.*, 2025). In the present study, melatonin application extended fruit shelf life by up to 9 days (Table 3). The treatment effectively delayed the loss of firmness, suppressed the decline in soluble solids and titratable acids and enhanced the fruit's antioxidant capacity, thereby contributing to prolonged freshness (Ze *et al.*, 2021; Hei *et al.*, 2025; Feng *et al.*, 2022). Comparable outcomes have been reported in apples (Onik *et al.*, 2021), mango (Njie *et al.*, 2022), sweet cherries (Carrión-Antolí *et al.*, 2022) and banana (Lin *et al.*, 2026). Melatonin plays an essential role in regulating fruit ripening through its interaction with ethylene and abscisic acid signalling pathways, thereby influencing the timing and synchrony of ripening while delaying senescence in fruit tissues (Aghdam *et al.*, 2024).

CONCLUSION

The findings of the study reveal that preharvest application of melatonin had a positive influence on maintaining the post-harvest quality of guava fruits by delaying ripening and reducing the rate of physiological and biochemical changes during storage. Treated fruits exhibited better retention of firmness and nutritional components, along with lower weight loss compared to untreated fruits. The regulation of sugar metabolism and antioxidant systems further contributed to improved fruit stability and slower senescence. Among the treatments, 100 µM melatonin was identified as the most effective concentration for preserving quality attributes and extending shelf life. In conclusion, exogenous melatonin represents a promising natural preservative that extends postharvest fruit shelf life by mitigating oxidative damage and strengthening antioxidant defences in guava.

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Informed consent

This study did not involve any experiments on human participants or animals. Therefore, ethics approval and permits were not required.

Conflict of interest

The authors declare that there are no conflicts of interest regarding the publication of this article.

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